

HISHAM UL HAKEEM A

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PROFESSIONAL SUMMARY

B.E. graduate in Computer Science (AI & ML) student with hands-on experience building Machine learning, RAG systems, CNN-based computer vision pipelines, and applied ML products across two internships. Seeking AI Engineer / Applied ML roles where I can contribute to production-grade AI systems and agentic workflows.

EDUCATION

B.E. Computer Science (AI & ML) — PES College of Engineering, Mandya

Dec 2022 – May 2026

12th Grade PCMC — ST. PHILOMENA'S PU COLLEGE

Jun 2020 – May 2022

EXPERIENCE

Artificial Intelligence Intern | Personifyfy, Mysore, India

May 2025 – Jul 2025

- Trained and optimized CNN-based image classification models for multi-class recognition tasks, fine-tuning a ResNet-50 transfer learning model to achieve 87% validation accuracy across 16 classes — a 40% relative improvement over the baseline CNN (62%).
- Built automated preprocessing and label encoding pipelines with standardized image resizing, normalization, and real-time augmentation (ImageDataGenerator), reducing training data preparation time by ~50% and improving model generalization.
- Designed controlled augmentation strategies (rotation, flipping, zoom, brightness shifts) that reduced model overfitting by improving validation accuracy stability across epochs.

Generative AI Intern | Learners Byte, Mysore, India

Jan 2026 – May 2026

- Architected a production-grade RAG healthcare assistant using LangChain, FAISS vector search, and Claude/OpenAI LLMs to analyze medical reports and generate patient-friendly summaries, reducing report interpretation time for end users by an estimated 60%.
- Engineered a hybrid document ingestion pipeline combining OCR and PDF parsing (pdfplumber, PyMuPDF) to automate structured extraction of clinical biomarkers from unstructured medical PDFs, processing reports with >95% field extraction accuracy.
- Implemented an automated biomarker anomaly detection layer that mapped extracted lab values against clinical reference ranges, flagging out-of-range results and exporting structured PDF summaries — eliminating the need for manual report review.

PROJECTS

Smart Attendance System | Python, OpenCV, ResNet, SSD, Flask, FAISS, SQLite

- Engineered an automated facial recognition attendance system using SSD-based face detection and 128-dimensional ResNet face embeddings stored in a FAISS index, achieving 84% recognition accuracy across diverse lighting conditions and camera angles.
- Built a Flask web dashboard with real-time attendance logging, CSV export, and anti-proxy detection logic, reducing manual attendance marking effort by 70–80% across enrolled student cohorts.
- Optimized the face recognition pipeline for stable inference under lighting and pose variation by fine-tuning embedding thresholds on enrolled image datasets, maintaining consistent F1 performance without retraining.

AI Resume Matcher | Python, Scikit-learn, NLP, TF-IDF, SpaCy, NLTK, Flask

- Developed an end-to-end ML-powered resume classification system using TF-IDF vectorization and Scikit-learn classifiers to predict relevant job categories from raw resume text, improving job-resume matching precision over keyword-based baselines.
- Applied NLP preprocessing pipelines (tokenization, stopword removal, lemmatization via SpaCy/NLTK) to normalize resume text into clean, structured feature vectors suitable for downstream classification.
- Deployed the trained model as a Flask web application with resume upload, real-time inference, and category visualization, making the system accessible to non-technical recruiters without API integration.

DermAI | Python, TensorFlow, Keras, CNN, OpenCV, Geolocation API

- Built a CNN-based dermatological image classifier trained on labeled skin disease datasets, achieving 73% validation accuracy across multiple disease categories through structured preprocessing, augmentation, and hyperparameter tuning.
- Integrated a geolocation-aware healthcare discovery feature using mapping APIs to surface nearby doctors, hospitals, and clinics based on user location — reducing time-to-consultation for users who received a positive preliminary screening.

TECHNICAL SKILLS

Languages: Python, SQL, HTML

ML / DL Frameworks: TensorFlow, Keras, PyTorch (basic), Scikit-learn, OpenCV, Hugging Face Transformers

GenAI & LLM Stack: LangChain, FAISS, RAG, vLLM, Embeddings, Prompt Engineering, Vector Databases

MLOps & Cloud: MLflow, AWS SageMaker, AWS Step Functions, Docker (basic), Git, Google Colab

APIs & Backend: FastAPI, REST APIs, MongoDB

Data & NLP: NumPy, Pandas, NLTK, SpaCy, TF-IDF, Matplotlib, Seaborn

Core Concepts: Machine Learning, Computer Vision, Deep Learning, NLP, Transfer Learning, Applied ML, Model Deployment

ACHIEVEMENTS

- Secured 2nd place among 42 competing teams at Prompt War, a timed AI prompt engineering competition — demonstrating practical LLM reasoning, prompt chaining, and creative problem-solving under constraint.
- Organized and led Prompt War 2.0 as the sole event director — designed the challenge framework, managed end-to-end logistics for 40+ participants, and created prompt-engineering tasks that evaluated multi-step reasoning and instruction-following ability.